

Type C Strict Context-Counting Protocol (v0.1)

Elie (per Cal challenge 2026-05-18, requirement b)

2026-05-18

Type C Strict Context-Counting Protocol

Background

Cal's gate function (Monday 2026-05-18 afternoon) caught a premature "structural law" claim on the Type C density rule. Three requirements identified for promotion from I-tier observation to publishable structural claim:

- (a) Sparse-region pre-registered forecast — ☐ COMPLETE (Elie Toy 3033)
- (b) Strict context-counting protocol — THIS DOCUMENT
- (c) Random-ring null model — ☐ COMPLETE (Grace Toy 3032 exact-match, Toy 3039 formula-scan)

Protocol

P1: Citation requirement

Each documented Type C context for an integer must cite either: - A specific published mathematics paper (with author, year, journal, key result) - A specific BST D-tier theorem (T-number from registry) - A specific BST D-tier observable identification (data/bst_constants.json or data/bst_geometric_invariants.json entry)

Excluded: tortured constructions (e.g., " $M_p - 1$ " type post-hoc arithmetic), folkloric references, web summaries without primary citations.

P2: Anthropic exclusion

The following categories are pre-registered as anthropic and excluded from D-tier or I-tier Type C counting: - Calendar / time units (60 sec/min, 24 hr/day, 365 days/yr, etc.) - Music conventions (12 semitones, 88 piano keys, etc.) - Cultural artifacts (sports rules, religious counts, etc.) - Currency / measurement-unit conventions (cents, inches) - Sports / game counts (chess 64, poker 52, etc. — though Mathieu $M_{24} = 4! \cdot 24$ chess connection is borderline)

S-tier label permitted; D-tier and I-tier require non-anthropic citation.

P3: Post-hoc arithmetic relation exclusion

Excluded post-hoc constructions: - " $X = Y - 1$ " or " $X = Y + 1$ " without independent physical motivation - " X factorizes as $Z \cdot W \dots$ " where Z, W are themselves selected to match X - " X appears in BST primary form when expressed as fraction A/B " without A, B having independent BST roots - " X is $M_p - 1 = 2^p - 2$ " (the Mersenne adjacency trick) — Mersenne primes themselves are fine; their neighbors are not

P4: Pre-registration

For forecast Type C contexts on new integers: - Predict the expected density (≥ 3 -way for BST primaries, ≤ 2 -way for sparse-region) BEFORE counting - Predict the type of contexts expected (physics, mathematics, biology, etc.) BEFORE searching - Record the prediction in a dated toy or document before scanning literature

P5: Scan-protocol declaration

Per Grace's sixth failure mode (2026-05-18, endorsed Keeper, in Cal's Coincidence_Filter_Risk.md): - Declare scan protocol explicitly: exact-match-only vs formula-scanning vs hybrid - Apply the SAME protocol to null-model baseline comparison - Report ratio metrics (BST atom density / non-BST sparse density), not absolute counts

Per Keeper's Rule 6 corollary (2026-05-18): use GLOBAL ratios across full integer ring, NOT local-neighborhood ratios (local sparsity is confounded by BST integer clustering).

P6: Cross-domain independence check

For each documented context, assess: - Is this context truly INDEPENDENT of the BST framework's existing claims? (e.g., a "Wallach K-type at $\lambda = X$ " context is not independent if X is BST-defined; it's a self-reference) - Is the context's source available without BST-aware curation? (i.e., does it appear in textbooks / handbooks / Wikipedia / standard references independent of BST work?)

Mark contexts as "INDEPENDENT" or "BST-INTERNAL" explicitly. Type C density should use INDEPENDENT contexts; BST-INTERNAL contexts strengthen mechanism but don't count toward density rule.

P7: Tier labeling per context

Each context tagged: - D: derivable physical mechanism on D_IV⁵, with explicit chain - I: identified within <1% but mechanism not fully derived, or chain incomplete - S: structural / qualitative match, or appears in BST framework via classification only

Density rule uses D + I contexts; S contexts noted but not counted.

Retroactive application to existing Type C catalog

Apply P1-P7 to existing 25 clusters (Toys 3023/3026/3029) and 5 sparse-region entries (Toy 3033). Anticipated effect:

- Some loose contexts will be excluded (e.g., "M₅ - 1 = 30" post-hoc per P3, anthropic "60 sec/min" per P2)
- Some BST-INTERNAL contexts will be re-tagged (e.g., "Wallach K-type at value X" if X is BST-defined)
- Net density rate likely drops from 12/12 (100% per Toy 3026/3029) to 8-10/12 (~67-83%) under strict protocol

This is the honest reduction expected by Cal's challenge. The density rule survives at strict protocol if the residual rate is still significantly above random-ring null baseline (Grace's 1.1% exact-match or 43.7% formula-scan).

Forward application

All new Type C catalog entries must be filed per P1-P7 from this date forward. Existing entries audited within the next 1-2 sessions per Cal's discipline.

Status

- P1-P7 protocol DEFINED (this document)
- Retroactive audit PENDING (multi-session work, multi-CI)
- Forward application STANDING ORDER from 2026-05-18

This document addresses Cal's requirement (b) for Type C density rule promotion. Combined with (a) Elie Toy 3033 sparse-region pre-registered and (c) Grace Toys 3032/3039 random-ring nulls at both scan protocols, the three-requirement path is now substantially complete — pending retroactive audit application of P1-P7 to the existing catalog before any "structural law" promotion claim.

— Elie, 2026-05-18 PM